

AKIRA — Complete Project Architecture & Evolution

Full Technical Specifications, Architecture & Operational Workflows (Phases 1 to 4)

System: AKIRA Intelligence Platform | Date: September 2026 | Status: Verified & Approved

1. EXECUTIVE PROJECT OVERVIEW & PHILOSOPHY

AKIRA (Autonomous Knowledge & Intelligence Real-time Aggregator) is a real-world intelligence aggregation platform built on fact corroboration, source transparency, anti-hallucination pipelines, and canonical event clustering.

- **Canonical Event vs Article Model:** Articles are publisher reports. Canonical Events are real-world occurrences. Multiple reports merge into 1 canonical event with corroborating source badges.
- **First-Class Regional Priority:** Tamil Nadu is treated as a top-tier primary region with dedicated infrastructure, public announcement, and district taxonomies alongside India and World.
- **Anti-Hallucination & No Fake Live Data:** Transparent timestamps (published_at vs discovered_at) ensure no fabricated 'BREAKING' or 'LIVE' tags.

2. PHASE 1: DATABASE & AUTHENTICATION FOUNDATIONS

Normal Working Phase: Users can register, log in with JWT sessions, manage profiles, and save bookmarked events. User saved data is strictly private and isolated.

Technical Phase: PostgreSQL/Supabase schema with UUID keys, Row-Level Security (RLS) policies, JWT middleware, in-memory repository fallbacks, and automated test migrations.

3. PHASE 2: FRONTEND MVP & UI FOUNDATION

Normal Working Phase: High-impact responsive dark UI featuring glassmorphism cards, cyan/indigo accent lighting, region selector tabs, category pills, live stream explorer, search drawers, and bookmark views.

Technical Phase: React 18, TypeScript, Vite, React Router v6, Lucide icons, TanStack React Query, and a handcrafted 100% Tailwind-free Vanilla CSS design system (index.css).

4. PHASE 3: CORE DATA LAYER & BACKEND FOUNDATION

Normal Working Phase: Connects UI to live Express REST API and PostgreSQL database, supporting pagination, region/category filters, search, and canonical event aggregation.

Technical Phase: Layered Express architecture (Routes -> Middleware -> Controllers -> Services -> Repositories -> Database), Zod request validation, sliding window rate limiter, safe HTML sanitizer, and internal key authentication.

5. PHASE 4: REAL NEWS INGESTION & EVENT PIPELINE

Normal Working Phase: Ingests live news from 14 vetted publishers (The Hindu, Indian Express, BBC, Reuters, TechCrunch, ScienceDaily, etc.). Automatically sanitizes, validates, deduplicates, classifies regions (Tamil Nadu priority), and clusters corroborating reports into single Canonical Events with transparent source badges.

Technical Phase: rss-parser with browser User-Agent, SSRF whitelist guard, 6,000ms timeout with exponential retry, tracking parameter stripper (urlNormalizer.ts), deterministic classifier (classifier.ts), 36h temporal & title token Jaccard event matcher (eventMatcher.ts), 100% idempotent PostgreSQL persistence, and background cron scheduler (scheduler.ts).

6. COMPREHENSIVE FILE & MODULE INVENTORY

Layer / Module	Key File Paths	Core Responsibility
Database & Seed	server/src/db/migrations/ 001_initial_schema.sql 002_rls_policies.sql 003_seed_data.sql neon_schema.sql	Core tables, RLS policies, foreign keys, and seed data for 14 sources & 15 categories.
Ingestion Pipeline	server/src/services/ingestion/ feedFetcher.ts, urlNormalizer.ts, articleValidator.ts, classifier.ts, eventMatcher.ts, scheduler.ts	Network fetching, timeout/retry, URL cleaning, HTML sanitizer, classification, clustering, background cron.
Repositories	server/src/repositories/ source, article, event, region, category .repository.ts	PostgreSQL queries, in-memory fallbacks, URL existence deduplication, and event-source attachment.
API Controllers & Middleware	server/src/controllers/ & server/src/middleware/ auth, rateLimit, validate, errorHandler	REST API endpoints, JWT auth, Zod query validation, sliding rate limiting, error logging.
Frontend UI	client/src/pages/ & client/src/components/ LiveTrending, Explore, EventDetail, AllNews, index.css	React 18 pages, glassmorphism design system, multi-source corroboration badges, and search.
Automated Tests	server/src/tests/ phase1.test.ts, phase3.test.ts, phase4.test.ts	Automated verification for Auth, RLS, REST APIs, Ingestion, Deduplication, and 100% Idempotency.

7. VERIFICATION RESULTS & SYSTEM STATUS

- **Phase 1 Test Suite:** 7/7 PASSED (Database migrations, JWT auth, private RLS isolation, sync guard)
- **Phase 3 Test Suite:** 14/14 PASSED (Repositories, REST APIs, Zod validation, XSS sanitization, rate limits)
- **Phase 4 Ingestion Suite:** 9/9 PASSED (URL normalization, validation, classifier, event clustering, 100% idempotency, health tracking)
- **Build Verification:** Client Vite production build PASSED (1,783 modules, 0 errors); Server TypeScript build PASSED (0 errors).